

BEE 4750 Lab 2: Monte Carlo

Name:

ID:

Due Date

Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

```
Activating project at `~/Documents/GitHub/lab-2-nar84-design`
```

```
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a “seed,” which controls the sequence of numbers which are generated (since these are not *really* random, just “pseudorandom”). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(67)
```

`TaskLocalRNG()`

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

Seeds and Reproducing Solutions

If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the [Distributions.jl package](#). The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

```
# you don't have to name this "normal_distribution"
#  is the mean and  is the standard deviation
normal_distribution = Normal( , )
# a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for ar
uniform_distribution = Uniform(a, b)
```

There are lots of both [univariate](#) and [multivariate](#) distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

```
# draw n samples
rand(normal_distribution, n)
```

Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a [Discrete Uniform distribution](#).

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

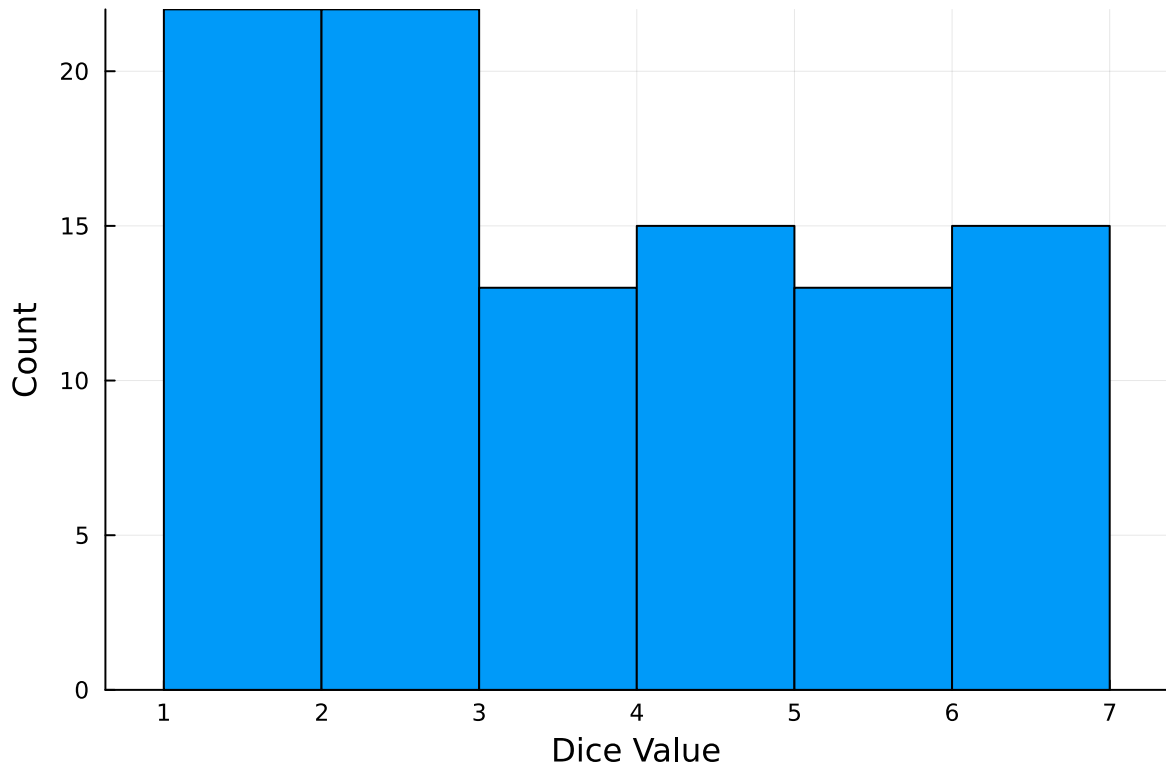
100-element Vector{Int64}:

```
6
6
1
4
3
2
1
2
1
6

3
1
6
3
6
5
1
1
1
```

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim \text{LogNormal}(1.2, 0.3)$;
2. $h \sim \text{GeneralizedExtremeValue}(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

```
L = 200000 #maximum loss in dollars
k = 0.8 #slope of the depth-damage relationship
h0 = 3 #inflection point

hLog = LogNormal(1.2, 0.3) #h value for lognormal

hLog_samples = rand(hLog, 1000000)

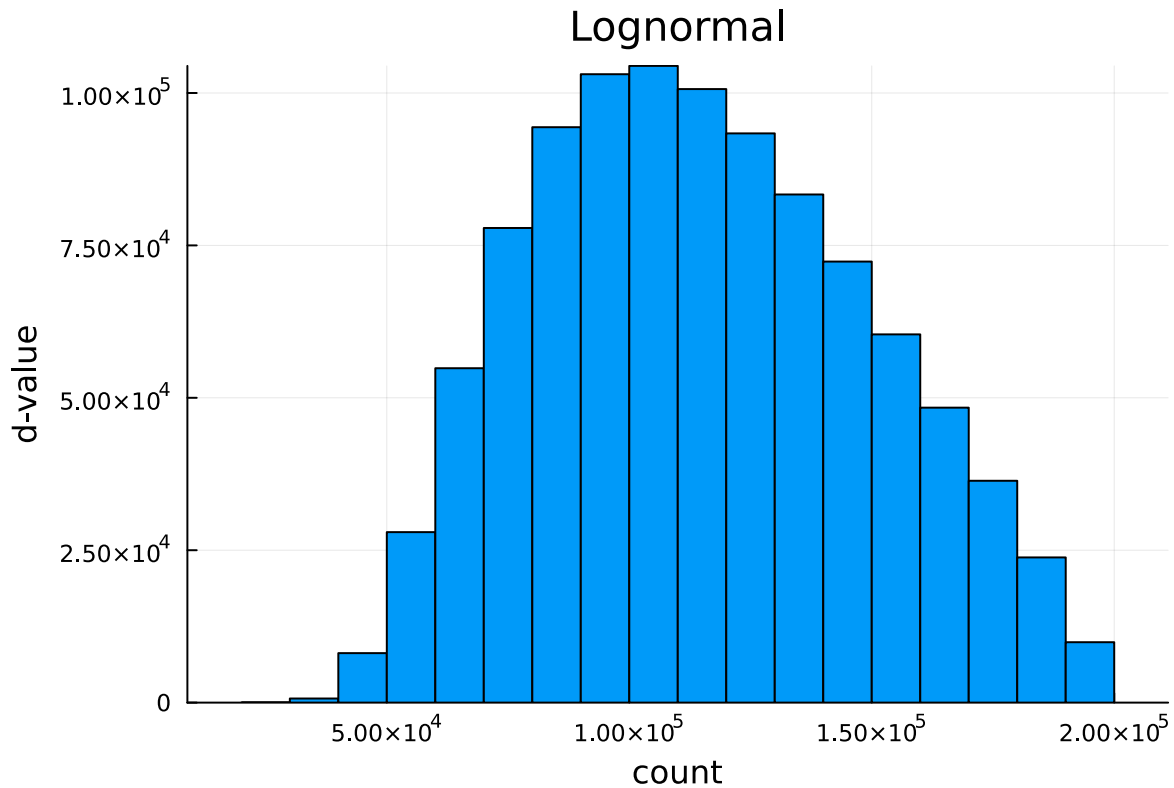
function indicator(array)
    if array > 0.0
        indicator = 1
    else
        indicator = 0
    end
    return indicator
end

function d(d) #d is a percentage
    return indicator(d) * L / (1 + exp.(-k * (d .- h0)))
end
```

```

log = d.(hLog_samples)
histogram(log, legend=:false, bins=20, title = "Lognormal")
ylabel!("d-value")
xlabel!("count")

```



```

L = 200000 #maximum loss in dollars
k = 0.8 #slope of the depth-damage relationship
h0 = 3 #inflection point

hGen = GeneralizedExtremeValue(3, 0.9, -0.15) #h value for lognormal

hGen_samples = rand(hGen, 1000000)

function indicator(array)
    if array > 0.0
        indicator = 1
    else
        indicator = 0
    end
end

```

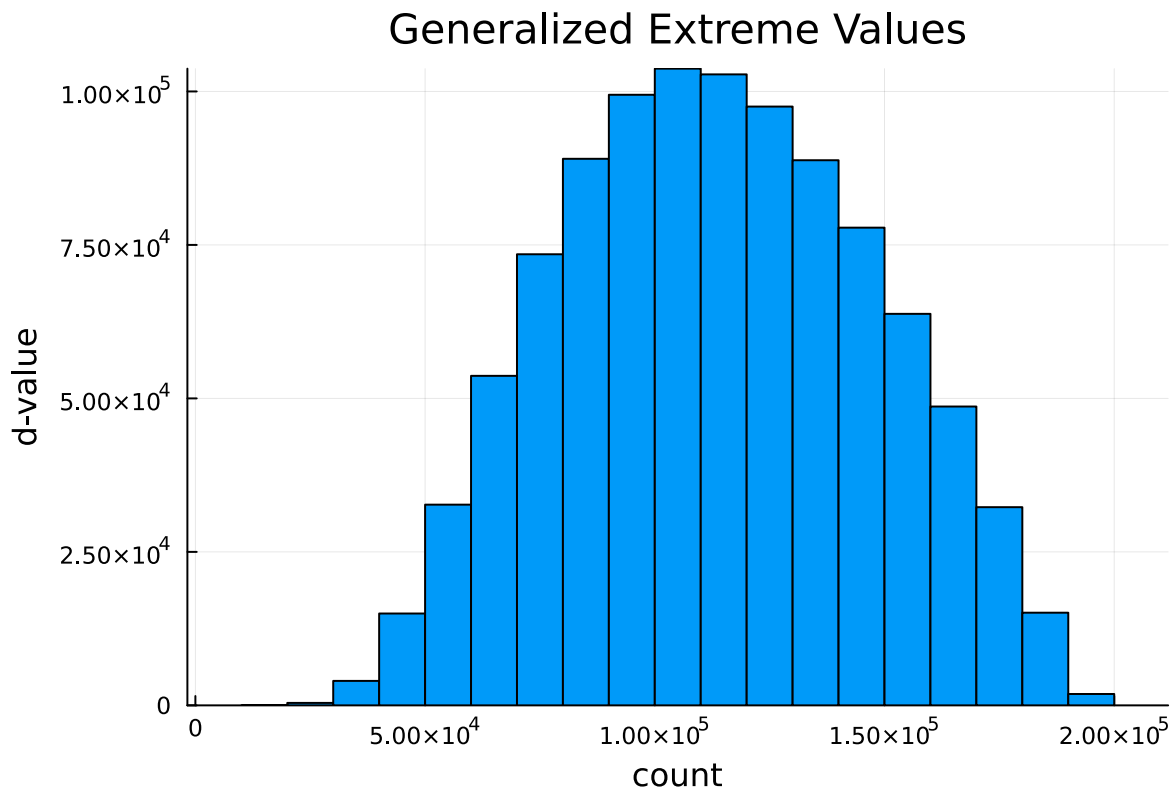
```

    return indicator
end

function d(d) #d is a percentage
    return indicator(d) * L / (1 + exp.(-k * (d .- h0)))
end

gen = d.(hGen_samples)
histogram(gen, legend=:false, bins=20, title = "Generalized Extreme Values")
ylabel!("d-value")
xlabel!("count")

```



```

L = 200000 #maximum loss in dollars
k = 0.8 #slope of the depth-damage relationship
h0 = 3 #inflection point

hLog = LogNormal(1.2, 0.3) #h value for lognormal
hLog_samples = rand(hLog, 1000000)

```

```

hGen = GeneralizedExtremeValue(3, 0.9, -0.15) #h value for lognormal

hGen_samples = rand(hGen, 1000000)

function indicator(array)
    if array > 0.0
        indicator = 1
    else
        indicator = 0
    end
    return indicator
end

function d(d) #d is a percentage
    return indicator(d) * L / (1 + exp.(-k * (d .- h0)))
end

log = d.(hLog_samples)
gen = d.(hGen_samples)

mean_gen = mean(gen)
mean_log = mean(log)
q99_log = quantile(log, 0.99)
q99_gen = quantile(gen, 0.99)

println("The mean loss for the lognormal distribution is $mean_log")
println("The mean loss for the generalized extreme value distribution is $mean_gen" )
println("The 99th percentile loss for the lognormal distribution is $q99_log")
println("The 99th percentile loss for the generalized extreme value distribution is $q99_gen")

```

```

The mean loss for the lognormal distribution is 115175.14478535965
The mean loss for the generalized extreme value distribution is 113492.0237854434
The 99th percentile loss for the lognormal distribution is 189844.2790325385
The 99th percentile loss for the generalized extreme value distribution is 183331.4118968733

```

With $n = 1,000,000$, there are enough samples so that the 1% isn't heavily influenced by outliers and change frequently with different seeds. Lower sample sizes can see greater differences in the tail depending on the set of random values. The estimates differed a little bit, but not by much. A change in order of magnitude would be significant, but this difference was only by about 2,000. The reason for the slight difference is because log normal distributions are not skewed, but GEVs are towards higher values. Because of this, the GEV is a little less since

it had many values of 0 when h was less than 1. Obtaining real data would be important to determine how the curves work. Maybe it is typically skewed towards one side. This should be reflected in the probability distribution in order to be more accurate.

References

Put any consulted sources here, including classmates you worked with/who helped you.

I worked with Sarah Vafiadis. I got help for wording from StackOverflow